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1. Write a program to sort the array elements using Merge Sort Technique.

Program:

```
#include <stdio.h>
```

```
void merge(int a[], int low, int mid, int high)
```

```
{
```

```
    int i = low, j = mid + 1, k = 0;
```

```
    int temp[100];
```

```
    while (i <= mid && j <= high)
```

```
    {
```

```
        if (a[i] < a[j])
```

```
            temp[k++] = a[i++];
```

```
        else
```

```
            temp[k++] = a[j++];
```

```
    }
```

```
    while (i <= mid)
```

```
        temp[k++] = a[i++];
```

```
    while (j <= high)
```

```
        temp[k++] = a[j++];
```

```
    for (i = low, k = 0; i <= high; i++, k++)
```

```
        a[i] = temp[k];
```

```
}
```

```
void mergeSort(int a[], int low, int high)
```

```
{
```

```
    if (low < high) {
```

```
        int mid = (low + high) / 2;
```

```

        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);
        merge(a, low, mid, high);
    }
}

int main()
{
    int a[100], n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    mergeSort(a, 0, n - 1);
    printf("Sorted array:\n");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}

```

Terminal Process:

```

● PS C:\Users\chand\OneDrive\DAA-2026> gcc merge.c -o mer
● PS C:\Users\chand\OneDrive\DAA-2026> ./mer
Enter number of elements: 5
Enter elements:
12
34
56
78
21
Sorted array:
12 21 34 56 78
○ PS C:\Users\chand\OneDrive\DAA-2026>

```

2. Write a program to sort the array elements using Bucket Sort Technique.

Program:

```
#include <stdio.h>

void bucketSort(int a[], int n)
{
    int bucket[100] = {0};

    int i, j = 0;
    for (i = 0; i < n; i++)
        bucket[a[i]]++;
    for (i = 0; i < 100; i++)
    {
        while (bucket[i] > 0)
        {
            a[j++] = i;
            bucket[i]--;
        }
    }
}

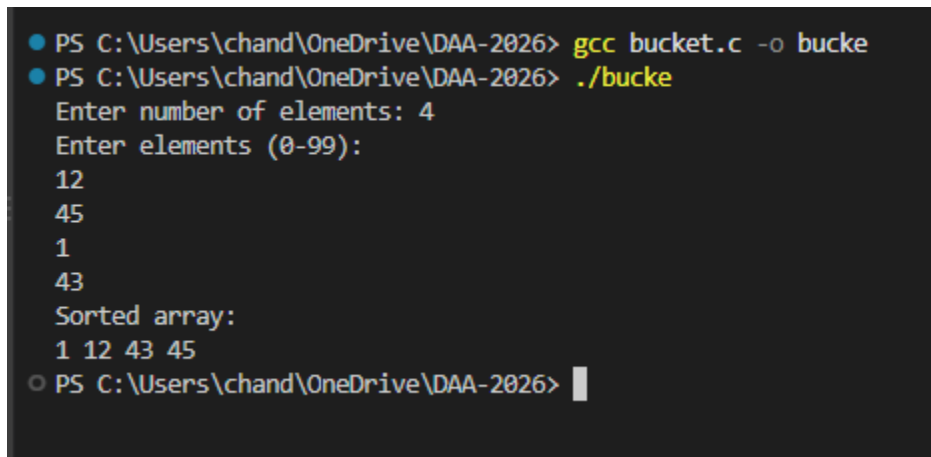
int main()
{
    int a[100], n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements (0-99):\n");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
}
```

```

    bucketSort(a, n);
    printf("Sorted array:\n");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}

```

Terminal Process:



```

PS C:\Users\chand\OneDrive\DAA-2026> gcc bucket.c -o bucke
PS C:\Users\chand\OneDrive\DAA-2026> ./bucke
Enter number of elements: 4
Enter elements (0-99):
12
45
1
43
Sorted array:
1 12 43 45
PS C:\Users\chand\OneDrive\DAA-2026>

```

3. Write a program to sort the array elements using Quick Sort Technique.

Program:

```

#include <stdio.h>

void swap(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}

int partition(int a[], int low, int high) {
    int pivot = a[high];
    int i = low - 1;

```

```

int j;
for (j = low; j < high; j++)
{
    if (a[j] < pivot)
    {
        i++;
        swap(&a[i], &a[j]);
    }
}
swap(&a[i + 1], &a[high]);
return i + 1;
}

void quickSort(int a[], int low, int high)
{ if (low < high){
    int pi = partition(a, low, high);
    quickSort(a, low, pi - 1);
    quickSort(a, pi + 1, high);
}
}

int main() { int a[100], n, i;

printf("Enter number of elements: ");
scanf("%d", &n);

printf("Enter elements:\n");
for (i = 0; i < n; i++)
    scanf("%d", &a[i]);

quickSort(a, 0, n - 1);

printf("Sorted array:\n");
for (i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;
}

```

Terminal Process:

```
● PS C:\Users\chand\OneDrive\DAA-2026> gcc quick.c -o quick
● PS C:\Users\chand\OneDrive\DAA-2026> ./quick
Enter number of elements: 5
Enter elements:
32
54
21
65
12
Sorted array:
12 21 32 54 65
○ PS C:\Users\chand\OneDrive\DAA-2026> |
```